

postdetector

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1 Compute detector signals in a post-processing mode

This code uses the *gpi_views* subroutine, e.g., as in the example *gpicamera.web* file to compute the end points of the chord representing each camera pixel as well as the contributions of each zone to that chord's signal. Since the *gpi_views* subroutine depends on data specific to the GPI application, it cannot be easily adapted to other situations. This version has been generalized in that it handles both D_2 (with photons coming from D, D_2 , and D_2^+) and He puffed gases and has been adapted to multiple GPI applications.

By default, *postdetector* runs with output summed over source groups. Output from an individual source group can be obtained by specifying its number on the command line:

```
postdetector 1
```

The only problem specific information appearing in this file are the camera resolution and the target plane. The data for the latter are hardcoded here with the specific applications identified with the *APP* macro. In principle, these could be moved to an input file. However, the camera resolution must remain in macro statements since the corresponding parameters appear in dimension statements. Note that all of the application specific information must match that in the corresponding *gpi_views* routine (see, e.g., [gpicaamera.web](#)). Note also that [gpicaamera.web](#) is to be used with the *APP = NSTX* option here. The other values of *APP* are paired with corresponding variants of the *detector_setup* routine in [gpicaamera.web](#) that are *not* provided with DEGAS 2.

In addition to the total emission rate (i.e., the quantity to be compared with the experimentally observed emission), various other quantities are computed to facilitate checking the simulated camera geometry (e.g., the “target” *R* and *Z*) and to facilitate interpretation of the results (e.g., the “effective densities”). Regarding the latter, no single quantity has been identified as the most useful. For examples, see D. P. Stotler et al., Contrib. Plasma Phys. **44**, 294 (2004) and J. R. Myra et al., Phys. Plasmas **18**, 012305 (2011).

These quantities and the corresponding file (or variable) names are:

Name	Description	Symbol
emtt	Total emission rate (W/(m ² sr))	S
emsp	Emission rate by species <i>p</i> (W/(m ² sr))	S_p
tarr	Target major radius (m)	$R(j_i)$
tarz	Target vertical coordinate (m)	$Z(j_i)$
tate	Target electron temperature (eV)	$T_e(j_i)$
tane	Target electron density (m ⁻³)	$n_e(j_i)$
fwpa	Emission rate per atom (W)	$F(j_i) = S(j_i)/n_1(j_i)$
nef1	Effective test density 1 (m ⁻² sr ⁻¹)	$S/F(j_i)$
favg	Average emission rate per atom (W)	$\langle F \rangle \equiv \langle S \rangle / \langle n_1 \rangle$
nef3	Effective test density 3 (m ⁻² sr ⁻¹)	$S/\langle F \rangle$
fwps	Specific emission rate per atom (W)	$S_1/n_1(j_i)$
nsf1	Specific effective test density 1 (m ⁻² sr ⁻¹)	$S_1/F(j_i)$
nsf3	Specific effective test density 3 (m ⁻² sr ⁻¹)	$S_1/\langle F_1 \rangle$
tnsp	Target density of species <i>p</i> (m ⁻³)	$n_p(j_i)$
n2sp	Effective density 2 of species <i>p</i> (m ⁻² sr ⁻¹)	$n_{\text{eff2},p}$
fasp	Average specific emission rate per species <i>p</i> (W)	$\langle F_p \rangle \equiv \langle S_p \rangle / \langle n_p \rangle$

Where:

- “Species *p*” refers to the *p*th species in the problem species list.
- If the “emission rate by species” tally is present, the emission rate and derived quantities can be divided into contributions from the emitting species (e.g., D, D₂⁺, D₂). One of these is designated as the “principal” emitter (“*p*=1” e.g., D).

- The function F is the “atomic physics” function for the principal emitter, i.e., the emission rate per atom evaluated at the local electron density and temperature from the input atomic physics tables (i.e., the density of the excited state relative to ground multiplied by the Einstein coefficient and energy of the transition).
- The “target” quantities are set first by identifying the zone (j_i) along each chord (i) closest to the target plane and then utilizing the corresponding quantity for that zone.
- The “average” quantities (e.g., $\langle S \rangle$) are averaged along a viewing chord over zones within a specified distance (d_{l0} in the code; e.g., 5 cm) of the target plane.
- The second effective density is the emitting particle’s density integrated along the entire viewing chord (i.e., computed just as is the total emission rate).
- These four characters in the file name will be followed by a string of the form **sg?** where ? represents the source group number, with **sg0** indicating the sum over all groups (default).

\$Id: 834c22f8502539519a9a1ba42d26f60361e1e6a6 **\$**

```
"postdetector.f" 1 ≡
@m FILE 'postdetector.web'

@m MPI_messages 0
@m CHORD_DATA 0
@m chord_stride 5
```

The main program.

```
"postdetector.f" 1.1 ≡
program postdetector

    implicit_none_f77
    mp_common
    so_common
    implicit_none_f90

    integer nargs, o_grp
    character*LINELEN arg

    mp_decls
    sy_decls
    st_decls

    mpi_init

    if (mpi_master)
        call readfilenames
    call degas_init    // In degasinit.web

    if (mpi_master) then    /* This is not MPI-aware. Instead, just need to extract the required data
        and broadcast it to the slaves. */
        call nc_read_output    // In pmimatread.web

        nargs = arg_count()
        if (nargs > 0) then
            assert(nargs ≡ 1)
            call command_arg(nargs, arg)
            o_grp = read_integer(arg)
            assert(so_check(o_grp))
        else
            o_grp = 0
        end if
        call fill_views_master(o_grp)
    else
        call fill_views_slave
    end if

    mpi_end

    stop
end
```

⟨ Functions and Subroutines 1.2 ⟩

The master routine for evaluating the views.

```
"postdetector.f" 1.2 ≡
  @m NSTX 0    // Macro for application. Could specify different
  @m CMOD_XPT 1    // nx and ny here if needed.
  @m CMOD_MID 2
  @m NSTX_2010 3
  @m NSTX_ENDD 4

  @m APP NSTX

@#if 0
@#if (APP ≡ NSTX_2010)
  @m nx 64
  @m ny 80
@#elif (APP ≡ NSTX_ENDD)
  @m nx 123    // Note that the actual camera pixel indices
  @m ny 56     // do not begin at 1.
@#else        // C-Mod or old NSTX
  @m nx 64
  @m ny 64
@#endif
@#else
  @m nx 5
  @m ny 5
@#endif
@m nsd 3    // Maximum number of test species yielding photons

  /* Remove characters causing file name problems */
@m name_clean(s, sp) sp = s
  ind_tmp = index(sp, '(')
  if (ind_tmp > 0)
    sp(ind_tmp : ind_tmp) = '_'
  ind_tmp = index(sp, ')')
  if (ind_tmp > 0)
    sp(ind_tmp : ind_tmp) = '_'
  ind_tmp = index(sp, '|')
  if (ind_tmp > 0)
    sp(ind_tmp : ind_tmp) = '_'
  ind_tmp = index(sp, '+')
  if (ind_tmp > 0)
    sp(ind_tmp : ind_tmp) = 'p'

⟨Functions and Subroutines 1.2⟩ ≡
  subroutine fill_views_master(o_grp)

    define_dimen(species_ind, nsd)

    define_varp(em_data, FLOAT, zone_ind)
    define_varp(em_sp_data, FLOAT, species_ind, zone_ind)
    define_varp(test_density, FLOAT, species_ind, zone_ind)
    define_varp(zone_frgs, FLOAT, zone_ind)

    implicit_none_f77
    sp_common
    pr_common
```

```

tl_common
ou_common
bk_common
zn_common
de_common
mp_common
@#if  $\neg$ HDF4
    po_common
    rf_common
@#endif
    implicit_none_f90

    integer o_grp     // Input

    integer zone, test, ix, iy, inverted_iy, algorithm, nviews, nfrag, nparcel, nslaves, j, nstart, num,
        slave, send_data, iview, close_zone, nx_peak, iy_peak, is, total_tally, species_tally, itally, len,
        ns, ind_tmp, igrp, inum, ifrag, this_grp, view_ptr
    integer index_parameters tl_index_max, target_map nx, ny, target_map_slave nx, ny
    real em_temp, em_rate, halfwidth, signal, a0_p, a1_p, a2_p, dist, min_dist, opt, max_opt, em_avg,
        a0, a1, a2, a3, a3_p, phi, d_l0, d_opt, peak, cy_l, cy_r
    real x_array nx, y_array ny, signal_array nx, ny, signal_array_slave nx, ny, temp_array nx, ny,
        target_r nx, ny, target_z nx, ny, target_n nx, ny, nsd, w_per_atom nx, ny, w_sp_per_atom nx, ny,
        n_eff1 nx, ny, n_sp_eff1 nx, ny, n_eff2 nx, ny, nsd, n_eff2_slave nx, ny, nsd, fwpa_avg nx, ny,
        fwpa_avg_slave nx, ny, n_eff3 nx, ny, n_sp_eff3 nx, ny, target_te nx, ny, target_ne nx, ny,
        signal_sp_array nx, ny, nsd, fwpa_sp_avg nx, ny, nsd, signal_sp_array_slave nx, ny, nsd,
        fwpa_sp_avg_slave nx, ny, nsd, profile ny, n_eff2_sum nsd, signal_sp nsd, n_avg nsd, em_sp_avg nsd
    character* sp_sy_len clean_sy
    character*1 s
    character*3 s_grp
    character*10 name_grp
    logical have_zone_frgs
@#if CHORD_DATA
    real r_zone, phi_zone, chord_length
    character*2 ixlab, iylab

    vc_decl(chord_vec)
@#endif
@#if  $\neg$ HDF4
    integer e, fileid
    character*FILELEN postdet_file

    po_ncdecl
    nc_decls
@#endif
    vc_decl(points de_view_start:de_view_end)
    external extract_output_datum, gpi_views
    real extract_output_datum

    declare_varp(em_data)
    declare_varp(em_sp_data)
    declare_varp(test_density)
    declare_varp(zone_frgs)

    (Memory allocation interface 0)
    st_decls

```

```

vc_decls
mp_decls
zn_decls
tl_decls

var_alloc(em_data)
var_alloc(em_sp_data)
var_alloc(test_density)
var_alloc(zone_frgs)

do zone = 1, zn_num
    em_data_zone = zero
    do is = 1, nsd
        em_sp_data_zone,is = zero
        test_density_zone,is = zero
    end do
end do

@#if  $\neg$ HDF4     // Use netCDF format

    po_nx = nx
    po_ny = ny
    po_num_dep_vars = 0
    var_realloca(po_xscale)
    var_realloca(po_yscale)
    var_realloca(po_image_data)
    var_realloca(po_image_labels)
    var_realloca(po_image_units)
    var_realloca(po_image_formats)
@#endif

    total_tally = int_undef
    species_tally = int_undef
    do itally = 1, tl_num
        len = string_length(tally_name_itally)
        if (len > 12) then
            if (tally_name_itally SP(len - 12 :)  $\equiv$  'emission_rate')
                total_tally = itally
            end if
            if (len > 23) then
                if (tally_name_itally SP(len - 23 :)  $\equiv$  'emission_rate_by_species')
                    species_tally = itally
                end if
            end if
        end do
        assert(tl_check(total_tally))
        if (tl_check(species_tally)) then
            ns = pr_test_num - 1     // Skip the "geometry" test species
        else
            ns = 1     // Only interested in one species; assume it's the first.
        end if

        em_temp = zero
        do zone = 1, zn_num
            if (zn_type(zone)  $\equiv$  zn_plasma) then
                index_parameters_tl_index_zone = zone
                if (o_grp  $\equiv$  0) then

```

```

    em_rate = extract_output_datum(index_parameters, 1, out_post_all, o_mean,
                                   tally_name_total_tally)
    s_grp = 'sg0'
    name_grp = 'all_groups'
else
    em_rate = extract_output_datum(index_parameters, 1, out_post_grp_o_grp,0,o_mean, o_mean,
                                   tally_name_total_tally)
    write(s_grp, '(a2,i1)') 'sg', o_grp
    write(name_grp, '(a8,i1)') 'src_grp', o_grp
end if
em_data_zone = em_rate
em_temp += em_rate * zn_volume(zone)
do test = 2, pr_test_num // Skip the "geometry" test species
    index_parameters_tl_index_test = test
    is = test - 1
    if (o_grp == 0) then
        test_density_zone,is = extract_output_datum(index_parameters, 1, out_post_all, o_mean,
                                                    'neutral_density')
    else
        test_density_zone,is = extract_output_datum(index_parameters, 1, out_post_grp_o_grp,0,o_mean,
                                                    o_mean, 'neutral_density')
    end if
    if (tl_check(species_tally)) then
        if (o_grp == 0) then
            em_rate = extract_output_datum(index_parameters, 1, out_post_all, o_mean,
                                           tally_name_species_tally)
        else
            em_rate = extract_output_datum(index_parameters, 1, out_post_grp_o_grp,0,o_mean,
                                           o_mean, tally_name_species_tally)
        end if
        em_sp_data_zone,is = em_rate
    end if
end do
end if
end do
@if MPI
    call MPI_bcast(em_data, zn_num, mpi_real, mpi_degas2_root, MPI_COMM_WORLD, mpi_err)
    call MPI_bcast(em_sp_data, zn_num*nsd, mpi_real, mpi_degas2_root, MPI_COMM_WORLD, mpi_err)
    call MPI_bcast(test_density, zn_num*nsd, mpi_real, mpi_degas2_root, MPI_COMM_WORLD, mpi_err)
    call MPI_bcast(ns, 1, mpi_int, mpi_degas2_root, MPI_COMM_WORLD, mpi_err)
@endif
if (em_temp > zero) then
    nviews = nx * ny
    have_zone_frgs = F
    igrp = int_unused
    if (de_grps > 0) then
        do igrp = 1, de_grps
            if (detector_num_views_igrp == nviews) then
                write(stdout, *) 'Using precomputed chord data for ', detector_name_igrp
                have_zone_frgs = T
                this_grp = igrp
            end if
        end do
    end if
end if

```

```

    end do
  end if    /* Copied this logic directly from do_flights_master. */
@#if MPI
  nslaves = mpi_size - 1    /* With precomputed zone_fragments, should just be running in serial mode.
    However, extending the code would only entail broadcasting the flag and the de class. */
  if (have_zone_fragments) then
    assert('Not set up for MPI with precomputed zone_fragments' ≡ '')
  end if
@#else
  nslaves = 0
@#endif
  nfrag = max(nviews / max(100, 10 * nslaves), 1)    // Size of each chunk
  nparcel = (nviews + nfrag - 1) / nfrag    // Number of chunks
@#if MPI
  do j = 0, min(nparcel, nslaves) - 1    // Send out the initial chunks
    nstart = j * nfrag
    num = min(nfrag, nviews - nstart)
    slave = j + 1
    @#if MPI_messages
      write(stderr, *) 'Send initial batch of ', num, ' to slave ', slave
    @#endif
    call slave_send_views(slave, nstart, num)
  end do
@#endif

  do j = min(nparcel, nslaves), nparcel - 1
    // Loop reading results and making additional requests
    nstart = j * nfrag
    num = min(nfrag, nviews - nstart)

    @#if MPI
      if (nslaves > 0) then
        call slave_receive_flag(slave)
        @#if MPI_messages
          write(stderr, *) 'Slave ', slave, ' is done with a batch'
          write(stderr, *) 'Sending ', nfrag, ' more to slave ', slave
        @#endif
        send_data = FALSE
        call slave_send_flag(slave, send_data)
        call slave_send_views(slave, nstart, num)
      else
        assert('Why is nslaves = 0 in an MPI run?' ≡ '')
      end if
    @#else
      < Compute Signal 2 >
    @#endif

  end do

@#if MPI
  do j = 0, min(nparcel, nslaves) - 1    // Retrieve data from slaves this time
    call slave_receive_flag(slave)
    send_data = TRUE
    call slave_send_flag(slave, send_data)
  end do
@#endif

```

```

call slave_receive_data(slave, signal_array_slave, target_map_slave, fupa_avg_slave,
    n_eff2_slave, signal_sp_array_slave, fupa_sp_avg_slave)

do iy = 1, ny
  do ix = 1, nx
    if (signal_array_slaveix,iy ≠ real_unused) then
      signal_arrayix,iy = signal_array_slaveix,iy
      assert(target_map_slaveix,iy ≠ int_unused)
      target_mapix,iy = target_map_slaveix,iy
      fupa_avgix,iy = fupa_avg_slaveix,iy
      do is = 1, ns
        assert(n_eff2_slaveix,iy,is ≠ real_unused)
        n_eff2ix,iy,is = n_eff2_slaveix,iy,is
        signal_sp_arrayix,iy,is = signal_sp_array_slaveix,iy,is
        fupa_sp_avgix,iy,is = fupa_sp_avg_slaveix,iy,is
      end do
    end if
  end do
end do

do j = 0, nslaves - 1    // Kill off slaves
  slave = j + 1
  call slave_send_views(slave, 0, 0)
end do
@#endif

do ix = 1, nx
  x_arrayix = areal(ix - 1)
end do

do iy = 1, ny
  @#if (APP ≡ NSTX_2010 ∨ APP ≡ CMOD_MID)
    y_arrayiy = areal(iy - 1)
  @#else
    inverted_iy = ny - iy + 1
    y_arrayinverted_iy = areal(iy - 1)
  @#endif
end do

call wrdatap(signal_array, x_array, y_array, 1, nx, 1, ny,
    'total_emission_rate' || name_grp, 'W/(m2_sr)', 'emtt' || s_grp, 'E11.3')

if (ns > 1) then
  do is = 1, ns
    do iy = 1, ny
      do ix = 1, nx
        temp_arrayix,iy = signal_sp_arrayix,iy,is
      end do
    end do
    name_clean(sp_sy(pr_test(is + 1)), clean_sy)    // see geomtesta.web
    assert(ns < 10)
    write(s, '(i1)') is
    call wrdatap(temp_array, x_array, y_array, 1, nx, 1, ny,
        'emission_rate_by' || trim(clean_sy) || '_' || name_grp, 'W/(m2_sr)',
        'ems' || s || s_grp, 'E11.3')
  end do
end if

```

```

        end do
    end if

@#if 0
    nx_peak = 33    // I.e., at cx = 32
    iy_peak = 0
    peak = zero
    do iy = 1, ny
        profile_iy = signal_array_nx_peak,iy
        if (profile_iy > peak) then
            peak = profile_iy
            iy_peak = iy
        end if
    end do
    assert(peak > zero)
    assert(iy_peak > 0)

    assert(half * peak > profile_1)
    cy_l = zero
    cy_r = zero
    do iy = 1, ny
        if ((cy_l == zero) ^ (profile_iy > half * peak)) then
            cy_l = y_array_iy-1 - (half * peak - profile_iy-1) / (profile_iy - profile_iy-1)
        end if
        if ((cy_r == 0) ^ (cy_l > zero) ^ (profile_iy < half * peak)) then
            cy_r = y_array_iy-1 - (half * peak - profile_iy-1) / (profile_iy - profile_iy-1)
        end if
    end do
    open(unit = diskout, file = 'detectorprofile.txt', status = 'unknown')
    write(diskout, *) 'At_cx=', x_array_nx_peak, ', peak=', peak, ', at_cy=', y_array_iy_peak
    write(diskout, *) 'Half_peak_at_cy=', cy_l, ', and ', cy_r, ', FWHM=', cy_r - cy_l
    write(diskout, *) 'iy_profile'
    do iy = 1, ny
        write(diskout, *) y_array_iy, profile_iy
    end do
    close(unit = diskout)
@#endif

    do iy = 1, ny
        do ix = 1, nx
            if (zn_check(target_map_ix,iy)) then
                zone = target_map_ix,iy
                target_r_ix,iy = sqrt(zone_center_zone,1^2 + zone_center_zone,2^2)
                target_z_ix,iy = zone_center_zone,3
                target_te_ix,iy = bk_temp(1, zone) / electron_charge
                target_ne_ix,iy = bk_n(1, zone)
                do is = 1, ns
                    target_n_ix,iy,is = test_density_zone,is
                end do
                if (test_density_zone,1 > zero) then
                    w_per_atom_ix,iy = em_data_zone / test_density_zone,1 /* This is just the emission by the
                        principal species in the multi-species case. Skip the other permutations. */
                    w_sp_per_atom_ix,iy = em_sp_data_zone,1 / test_density_zone,1
                else

```

```

        w_per_atomix,iy = zero
        w_sp_per_atomix,iy = zero
    end if
else
    target_rix,iy = zero
    target_zix,iy = zero
    target_teix,iy = zero
    target_neix,iy = zero
    do is = 1, ns
        target_nix,iy,is = zero
    end do
    w_per_atomix,iy = zero
    w_sp_per_atomix,iy = zero
end if
if (w_per_atomix,iy > zero) then
    n_eff1ix,iy = signal_arrayix,iy / w_per_atomix,iy
else
    n_eff1ix,iy = zero
end if
if (w_sp_per_atomix,iy > zero) then
    n_sp_eff1ix,iy = signal_sp_arrayix,iy,1 / w_sp_per_atomix,iy
else
    n_sp_eff1ix,iy = zero
end if
if (f_wpa_avgix,iy > zero) then
    n_eff3ix,iy = signal_arrayix,iy / f_wpa_avgix,iy
end if
if (f_wpa_sp_avgix,iy,1 > zero) then
    n_sp_eff3ix,iy = signal_sp_arrayix,iy,1 / f_wpa_sp_avgix,iy,1
else
    n_sp_eff3ix,iy = zero
end if
end do
end do
end if

call wrdatap(target_r, x_array, y_array, 1, nx, 1, ny, 'target_major_radius', 'm',
'tarr' || s_grp, 'E11.3')
call wrdatap(target_z, x_array, y_array, 1, nx, 1, ny, 'target_vertical_coord', 'm',
'tarz' || s_grp, 'E11.3')
call wrdatap(target_te, x_array, y_array, 1, nx, 1, ny, 'target_electron_temperatu\
re', 'eV', 'tate' || s_grp, 'E11.3')
call wrdatap(target_ne, x_array, y_array, 1, nx, 1, ny, 'target_electron_density', 'm^3',
'tane' || s_grp, 'E11.3')
call wrdatap(w_per_atom, x_array, y_array, 1, nx, 1, ny,
'emission_rate_per_atom' || name_grp, 'W', 'f_wpa' || s_grp, 'E11.3')
call wrdatap(n_eff1, x_array, y_array, 1, nx, 1, ny, 'effective_test_density_1\
' || name_grp, 'm^2_sr^-1', 'nef1' || s_grp, 'E11.3')
call wrdatap(f_wpa_avg, x_array, y_array, 1, nx, 1, ny, 'avg_emission_rate_per_ato\
m' || name_grp, 'W', 'favg' || s_grp, 'E11.3')
call wrdatap(n_eff3, x_array, y_array, 1, nx, 1, ny, 'effective_test_density_3\
' || name_grp, 'm^2_sr^-1', 'nef3' || s_grp, 'E11.3')

```

```

if (tl_check(species_tally)) then
  call wrdatap(w_sp_per_atom, x_array, y_array, 1, nx, 1, ny, 'specific_emission_rate_per\
    r_atom' || name_grp, 'W', 'fwps' || s_grp, 'E11.3')
  call wrdatap(n_sp_eff1, x_array, y_array, 1, nx, 1, ny, 'specific_effective_test_d\
    ensity_1' || name_grp, 'm^2_sr^-1', 'nsf1' || s_grp, 'E11.3')
  call wrdatap(n_sp_eff3, x_array, y_array, 1, nx, 1, ny, 'specific_effective_test_d\
    ensity_3' || name_grp, 'm^2_sr^-1', 'nsf3' || s_grp, 'E11.3')
end if

do is = 1, ns
  name_clean(sp_sy(pr_test(is + 1)), clean_sy)
  write(s, '(i1)') is
  do iy = 1, ny
    do ix = 1, nx
      temp_arrayix, iy = target_nix, iy, is
    end do
  end do
  call wrdatap(temp_array, x_array, y_array, 1, nx, 1, ny,
    'target_density_of' || trim(clean_sy) || '_' || name_grp, 'm^3', 'tns' || s || s_grp,
    'E11.3')
  do iy = 1, ny
    do ix = 1, nx
      temp_arrayix, iy = n_eff2ix, iy, is
    end do
  end do
  call wrdatap(temp_array, x_array, y_array, 1, nx, 1, ny,
    'effective_density_2_of' || trim(clean_sy) || '_' || name_grp, 'm^2_sr^-1',
    'n2s' || s || s_grp, 'E11.3')
  do iy = 1, ny
    do ix = 1, nx
      temp_arrayix, iy = fupa_sp_avgix, iy, is
    end do
  end do
  call wrdatap(temp_array, x_array, y_array, 1, nx, 1, ny, 'avg_specific_emission_rat\
    e_per' || trim(clean_sy) || '_' || name_grp, 'W', 'fas' || s || s_grp, 'E11.3')
end do

var_free(em_data)
var_free(em_sp_data)
var_free(test_density)
var_free(zone_frgs)
@#if ¬HDF4
  var_reallocb(po_xscale)
  var_reallocb(po_yscale)
  var_reallocb(po_image_data)
  var_reallocb(po_image_labels)
  var_reallocb(po_image_units)
  var_reallocb(po_image_formats)
  e = index(filenames_arrayoutputfile, '.nc') - 1
  assert(e > 0)
  postdet_file = filenames_arrayoutputfile SP(: e) || '_post.nc'

  fileid = nccreate(postdet_file, NC_CLOBBER, nc_stat)
  po_ncdef(fileid)

```

```
    call ncendef(fileid, nc_stat)  
    po_ncwrite(fileid)  
    call ncclose(fileid, nc_stat)  
    var_free(po_xscale)  
    var_free(po_yscale)  
    var_free(po_image_data)  
    var_free(po_image_labels)  
    var_free(po_image_units)  
    var_free(po_image_formats)  
@#endif  
  
    return  
end
```

See also sections 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, and 2.7.

This code is used in section 1.1.

2 Compute signal

This code is repeated in the master and slave *fill_views* routines.

```

⟨ Compute Signal 2 ⟩ ≡
    /* Specific parameters for this application */
    @#if (APP ≡ NSTX)
        a0 = -const(3.60752)    // Coefficients of the camera's "target plane"
        a1 = const(2.11865)
        a2 = const(3.17544)
        a3 = -one
        phi = const(6.146, 1) * PI / const(1.8, 2)    // Toroidal shift to my system
        d_l0 = const(5., -2)    // Range for averaging about target plane
        d_opt = const(2.5, -1)  // Range for searching for a close zone
    @#elif (APP ≡ CMOD.MID)
        a0 = -sin(const(-18.48) * PI / const(1.8, 2))    // A vertical plane at this
        a1 = cos(const(-18.48) * PI / const(1.8, 2))    // toroidal angle
        a2 = zero
        a3 = zero
        phi = zero    // No additional toroidal shift needed
        d_l0 = const(5., -2)
        d_opt = const(3.0, -1)
    @#elif (APP ≡ CMOD.XPT)
        a0 = -sin(const(33.0) * PI / const(1.8, 2))    // Again, a vertical plane
        a1 = cos(const(33.0) * PI / const(1.8, 2))
        a2 = zero
        a3 = zero
        phi = zero
        d_l0 = const(5., -2)
        d_opt = const(3.0, -1)
    @#elif (APP ≡ NSTX_2010)
        a0 = -const(3.58653)    // Coefficients of the camera's "target plane"
        a1 = const(2.08796)
        a2 = const(3.18466)
        a3 = -one
        phi = const(6.146, 1) * PI / const(1.8, 2)    // Toroidal shift to my system
        d_l0 = const(5., -2)    // Range for averaging about target plane
        d_opt = const(2.5, -1)  // Range for searching for a close zone
    @#elif (APP ≡ NSTX.ENDD)
        a0 = -sin(const(155.0) * PI / const(1.8, 2))    // Again, a vertical plane
        a1 = cos(const(155.0) * PI / const(1.8, 2))
        a2 = zero
        a3 = zero
        phi = zero
        d_l0 = const(5., -2)
        d_opt = const(3.0, -1)
    @#else
        assert('Target_plane_parameters_not_specified' ≡ '␣')
    @#endif

    /* Rotate Ricky's "target plane" to my coordinate system. The "p" denotes "prime". */
    a0_p = a0 * cos(phi) + a1 * sin(phi)
    a1_p = a1 * cos(phi) - a0 * sin(phi)

```

```

a2_p = a2
a3_p = a3
do iview = nstart, nstart + num - 1
  if (mod(iview + 1, nx) ≠ 0) then
    iy = (iview + 1) / nx + 1
    ix = (iview + 1) - (iy - 1) * nx
  else
    iy = (iview + 1) / nx
    ix = nx
  end if
  if (have_zone_fragments) then
    assert((this_grp ≥ 1) ∧ (this_grp ≤ de_grps))
    do zone = 1, zn_num
      zone_fragments_zone = zero
    end do
    view_ptr = de_view_pointer(iview + 1, this_grp)
    do inum = 1, de_zone_fragments_num_view_ptr
      ifrag = de_zone_fragments_start_view_ptr + (inum - 1)
      zone = de_zone_fragments_zones_ifrag
      assert(zn_check(zone))
      zone_fragments_zone = de_zone_fragments_ifrag
    end do
  else
    call gpi_views(ix, nx, iy, ny, vc_args(points_de_view_start), halfwidth, algorithm, zone_fragments)
  end if
@#if CHORD_DATA
  if ((mod(ix, chord_stride) ≡ 0) ∧ (mod(iy, chord_stride) ≡ 0)) then
    write(ixlab, '(i2.2)') ix
    write(iylab, '(i2.2)') iy
    open(unit = diskout, file = 'gpi_chord_' || ixlab || '_' || iylab, status = 'unknown')
    write(diskout, *) 'zonezone_fragdistanceRphizone_data_densities'
  end if
@#endif
  signal = zero
  em_avg = zero
  do is = 1, ns
    n_eff2_sum_is = zero
    signal_sp_is = zero
    n_avg_is = zero
    em_sp_avg_is = zero
  end do
  min_dist = const(1., 16)
  max_opt = const(1., -16)
  close_zone = int_undef
  do zone = 1, zn_num
@#if CHORD_DATA
    if ((zone_fragments_zone > zero) ∧ (mod(ix, chord_stride) ≡ 0) ∧ (mod(iy, chord_stride) ≡ 0)) then
      r_zone = sqrt(zone_center_zone,12 + zone_center_zone,22)
      phi_zone = atan2(zone_center_zone,2, zone_center_zone,1)
      vc_difference(zone_center_zone, points_de_view_start, chord_vec)
      chord_length = vc_abs(chord_vec)
      write(diskout, '(2x,i7,2x,1p,e16.8,0p,4(2x,f11.8),1p,4(2x,e16.8))') zone,

```

```

    zone_fragszone * zn_volume(zone), chord_length, r_zone, phi_zone, zone_centerzone,3,
    em_datazone, (test_densityzone,is, is = 1, ns)
end if
@#endif
    signal += em_datazone * zone_fragszone * zn_volume(zone)
do is = 1, ns
    n_eff2_sumis += test_densityzone,is * zone_fragszone * zn_volume(zone)
    signal_spis += em_sp_datazone,is * zone_fragszone * zn_volume(zone)
end do /* Hack to distill out zone closest to target plane. */
dist = abs(a0_p * zone_centerzone,1 + a1_p * zone_centerzone,2 + a2_p * zone_centerzone,3 + a3_p) /
    sqrt(a0_p2 + a1_p2 + a2_p2)
if (dist ≠ zero) then
    opt = zone_fragszone * zn_volume(zone) / dist
else /* In the C-Mod problems, can actually get dist = 0. Use a finite length for computing
    opt to distinguish these in-target-plane zones. */
    opt = zone_fragszone * zn_volume(zone) / const(1., -8)
end if
@#if 0
    if ((zone_fragszone > zero) ∧ (dist < min_dist)) then
@#endif
    if ((opt > max_opt) ∧ (dist ≤ d_opt)) then
        max_opt = opt
        min_dist = dist
        close_zone = zone
    end if /* Sums restricted to region around target plane. */
    if (dist ≤ d_l0) then
        em_avg += em_datazone * zone_fragszone * zn_volume(zone)
        do is = 1, ns
            n_avgis += test_densityzone,is * zone_fragszone * zn_volume(zone)
            em_sp_avgis += em_sp_datazone,is * zone_fragszone * zn_volume(zone)
        end do
    end if
end do
@#if CHORD_DATA
    if ((mod(ix, chord_stride) ≡ 0) ∧ (mod(iy, chord_stride) ≡ 0))
        close(unit = diskout)
    end if
@#endif
@#if (APP ≡ NSTX_2010 ∨ APP ≡ NSTX_ENDD ∨ CMOD_MID)
    inverted_iy = iy
@#else
    inverted_iy = ny - iy + 1
@#endif
/* Have a case now in which a chord does not reach the target plane. Will let close_zone remain
int_undef and then assign zero values to the associated target parameters. */
@#if 0
    assert(close_zone ≠ int_undef)
    assert(min_dist < const(1., 16))
@#endif
    target_mapix,inverted_iy = close_zone
    signal_arrayix,inverted_iy = signal
    if (n_avg1 > zero) then
        fwpa_avgix,inverted_iy = em_avg / n_avg1
    end if
end if

```

```

else
   $f_{wpa\_avg_{ix, inverted\_iy}} = zero$ 
end if
do  $is = 1, ns$ 
   $n\_eff2_{ix, inverted\_iy, is} = n\_eff2\_sum_{is}$ 
   $signal\_sp\_array_{ix, inverted\_iy, is} = signal\_sp_{is}$ 
  if ( $n\_avg_{is} > zero$ ) then
     $f_{wpa\_sp\_avg_{ix, inverted\_iy, is}} = em\_sp\_avg_{is} / n\_avg_{is}$ 
  else
     $f_{wpa\_sp\_avg_{ix, inverted\_iy, is}} = zero$ 
  end if
end do
end do

```

This code is used in sections 1.2 and 2.1.

Slave version for filling views.

⟨ Functions and Subroutines 1.2 ⟩ +=

@#if *MPI*

subroutine *fill_views_slave*

define_varp(em_data, FLOAT, zone_ind)

define_varp(em_sp_data, FLOAT, species_ind, zone_ind)

define_varp(test_density, FLOAT, species_ind, zone_ind)

define_varp(zone_frgs, FLOAT, zone_ind)

implicit_none_f77

mp_common

zn_common

de_common

implicit_none_f90

integer *iy, ix, tag, num, nstart, iview, is, ns, algorithm, close_zone, zone, inverted_iy, done_flag,*
send_data, inum, ifrag

integer *target_map*_{*nx,ny*}

real *halfwidth, signal, a0_p, a1_p, a2_p, dist, min_dist, a0, a1, a2, a3, phi, d_l0, d_opt, a3_p,*
em_avg, opt, max_opt

real *signal_array*_{*nx,ny*}, *n_eff2*_{*nx,ny,nsd*}, *fwpa_avg*_{*nx,ny*}, *signal_sp_array*_{*nx,ny,nsd*},
*fwpa_sp_avg*_{*nx,ny,nsd*}, *n_eff2_sum*_{*nsd*}, *signal_sp*_{*nsd*}, *n_avg*_{*nsd*}, *em_sp_avg*_{*nsd*}

logical *have_zone_frgs*

@#if *CHORD_DATA*

real *r_zone, phi_zone, chord_length*

character*2 *ixlab, iylab*

vc_decl(chord_vec)

@#endif

*vc_decl(points*_{*de_view_start:de_view_end*}*)*

declare_varp(em_data)

declare_varp(em_sp_data)

declare_varp(test_density)

declare_varp(zone_frgs)

⟨ Memory allocation interface 0 ⟩

vc_decls

mp_decls

zn_decls

var_alloc(em_data)

var_alloc(em_sp_data)

var_alloc(test_density)

var_alloc(zone_frgs)

call *MPI_bcast(em_data, zn_num, mpi_real, mpi_degas2_root, MPI_COMM_WORLD, mpi_err)*

call *MPI_bcast(em_sp_data, zn_num*nsd, mpi_real, mpi_degas2_root, MPI_COMM_WORLD, mpi_err)*

call *MPI_bcast(test_density, zn_num*nsd, mpi_real, mpi_degas2_root, MPI_COMM_WORLD, mpi_err)*

call *MPI_bcast(ns, 1, mpi_int, mpi_degas2_root, MPI_COMM_WORLD, mpi_err)*

do *iy = 1, ny*

do *ix = 1, nx*

*signal_array*_{*ix,iy*} = *real_unused*

*target_map*_{*ix,iy*} = *int_unused*

```

      fwpavgix,iy = real_unused
      do is = 1, ns
        n_eff2ix,iy,is = real_unused
        signal_sp_arrayix,iy,is = real_unused
        fwpavg_spix,iy,is = real_unused
      end do
    end do
  end do

loop: continue
  tag = 100
  call MPI_recv(num, 1, mpi_int, mpi_degas2_root, tag, MPI_COMM_WORLD, mpi_status, mpi_err)
  if (num == 0)
    goto break
  tag++
  call MPI_recv(nstart, 1, mpi_int, mpi_degas2_root, tag, MPI_COMM_WORLD, mpi_status, mpi_err)
  /* See comment in master routine. */
  have_zone_fragments = F
  < Compute Signal 2 >

  done_flag = TRUE
  tag = 200
  call MPI_send(done_flag, 1, mpi_int, mpi_degas2_root, tag, MPI_COMM_WORLD, mpi_err)

  tag = 300
  call MPI_recv(send_data, 1, mpi_int, mpi_degas2_root, tag, MPI_COMM_WORLD, mpi_status,
    mpi_err)
  if (send_data == FALSE) then
    goto loop
  else if (send_data == TRUE) then

    tag = 400

    call MPI_send(signal_array, nx * ny, mpi_real, mpi_degas2_root, tag, MPI_COMM_WORLD,
      mpi_err)
    tag = 401
    call MPI_send(target_map, nx * ny, mpi_int, mpi_degas2_root, tag, MPI_COMM_WORLD, mpi_err)
    tag = 402
    call MPI_send(fwpavg, nx * ny, mpi_real, mpi_degas2_root, tag, MPI_COMM_WORLD, mpi_err)
    tag = 403
    call MPI_send(n_eff2, nx * ny * nsd, mpi_real, mpi_degas2_root, tag, MPI_COMM_WORLD, mpi_err)
    tag = 404
    call MPI_send(signal_sp_array, nx * ny * nsd, mpi_real, mpi_degas2_root, tag,
      MPI_COMM_WORLD, mpi_err)
    tag = 405
    call MPI_send(fwpavg_sp, nx * ny * nsd, mpi_real, mpi_degas2_root, tag,
      MPI_COMM_WORLD, mpi_err)
    do iy = 1, ny
      do ix = 1, nx
        signal_arrayix,iy = real_unused
        target_mapix,iy = int_unused
        fwpavgix,iy = real_unused
        do is = 1, ns
          n_eff2ix,iy,is = real_unused
          signal_sp_arrayix,iy,is = real_unused

```

```

                fupa_sp_avgix, iy, is = real_unused
            end do
        end do
    end do

    goto loop
else
    assert('Illegal_value_of_send_data' ≡ ' ')
end if

break: continue

    var_free(em_data)
    var_free(em_sp_data)
    var_free(test_density)
    var_free(zone_fragments)
    return
end
@#else

    subroutine fill_views_slave    // dummy
        return
    end
@#endif

```

Send data to slave.

⟨ Functions and Subroutines 1.2 ⟩ +≡

```

@#if MPI
    subroutine slave_send_views(slave, nstart, num)
        implicit none_f77
        mp_common
        implicit none_f90
        mp_decls
        integer slave, nstart, num
        integer tag

        ⟨ Memory allocation interface 0 ⟩

        tag = 100
        call MPI_send(num, 1, mpi_int, slave, tag, MPI_COMM_WORLD, mpi_err)
        if (num ≡ 0)
            goto break
        tag++
        call MPI_send(nstart, 1, mpi_int, slave, tag, MPI_COMM_WORLD, mpi_err)

    break: continue
        return
    end
@#endif

```

Receive notice from a slave indicating that it is done with current batch.

⟨ Functions and Subroutines 1.2 ⟩ +≡

```
@#if MPI
  subroutine slave_receive_flag(slave)
    implicit none f77
    mp_common
    implicit none f90
    mp_decls
    integer slave
    integer tag, done_flag

    ⟨ Memory allocation interface 0 ⟩

    tag = 200
    call MPI_recv(done_flag, 1, mpi_int, MPI_ANY_SOURCE, tag, MPI_COMM_WORLD, mpi_status,
                  mpi_err)
    slave = mpi_status(MPI_SOURCE)
    assert(done_flag ≡ TRUE)
    return
  end
@#endif
```

Send a flag to a slave instructing it to send back data from its flights.

⟨ Functions and Subroutines 1.2 ⟩ +≡

```
@#if MPI
  subroutine slave_send_flag(slave, send_data)
    implicit none f77
    mp_common
    implicit none f90
    mp_decls
    integer slave, send_data    // Input
    integer tag                // Local

    ⟨ Memory allocation interface 0 ⟩

    tag = 300
    call MPI_send(send_data, 1, mpi_int, slave, tag, MPI_COMM_WORLD, mpi_err)

    return
  end
@#endif
```

Receive data from flights run by a slave.

⟨ Functions and Subroutines 1.2 ⟩ +≡

@#if MPI

```
subroutine slave_receive_data(slave, signal_array, target_map, fwpa_avg, n_eff2, signal_sp_array,
    fwpa_sp_avg)
    implicit none f77
    mp_common
    implicit none f90
    mp_decls
    integer slave
    integer target_map nx,ny
    real signal_array nx,ny, fwpa_avg nx,ny, n_eff2 nx,ny,nsd, signal_sp_array nx,ny,nsd,
        fwpa_sp_avg nx,ny,nsd
    integer tag
```

⟨ Memory allocation interface 0 ⟩

```
tag = 400
```

```
call MPI_recv(signal_array, nx * ny, mpi_real, MPLANY_SOURCE, tag, MPLCOMM_WORLD,
    mpi_status, mpi_err)
```

```
tag = 401
```

```
call MPI_recv(target_map, nx * ny, mpi_int, MPLANY_SOURCE, tag, MPLCOMM_WORLD,
    mpi_status, mpi_err)
```

```
tag = 402
```

```
call MPI_recv(fwpa_avg, nx * ny, mpi_real, MPLANY_SOURCE, tag, MPLCOMM_WORLD,
    mpi_status, mpi_err)
```

```
tag = 403
```

```
call MPI_recv(n_eff2, nx * ny * nsd, mpi_real, MPLANY_SOURCE, tag, MPLCOMM_WORLD,
    mpi_status, mpi_err)
```

```
tag = 404
```

```
call MPI_recv(signal_sp_array, nx * ny * nsd, mpi_real, MPLANY_SOURCE, tag,
    MPLCOMM_WORLD, mpi_status, mpi_err)
```

```
tag = 405
```

```
call MPI_recv(fwpa_sp_avg, nx * ny * nsd, mpi_real, MPLANY_SOURCE, tag,
    MPLCOMM_WORLD, mpi_status, mpi_err)
```

```
return
```

```
end
```

@#endif

Rearrange 2D array prior to writing. This configuration is specific to the 2004 NSTX Gas Puff Imaging experiments.

⟨ Functions and Subroutines 1.2 ⟩ +≡

```

subroutine wrdatap(pb2dat, xdat, ydat, kxmin, kx, kymin, ky, clabel, cunits, cfilt, cformt)
    implicit_none_f77
    implicit_none_f90

    integer kxmin, kx, ky, kymin    // Input
    character*(*) cunits, clabel, cfilt, cformt
    real xdatkx, ydatky, pb2datkx,ky

    integer ix, iy, ixprime, iyprime    // Local
    real xprimeny, yprimenx, pb2primeny,nx

    assert(kx ≡ nx)
    assert(ky ≡ ny)
@#if (APP ≡ NSTX)
    assert(kx ≡ ky)    /* The required mapping is equivalent to rowflip(transpose(q)) in Transform.
        Namely, swap ix and iy, and then invert the horizontal coordinate. Note that we just assume
        that the dimensions are the same in both directions. */
    do iy = kymin, ky
        do ix = kxmin, kx
            ixprime = iy
            iyprime = kx - ix + 1
            pb2primeixprime,iyprime = pb2datix,iy
        end do
    end do    /* Apply equivalent transformations to the scales. */
    do ix = kxmin, kx
        xprimeix = ydatix
    end do

    do iy = kymin, ky
        ixprime = ky - iy + 1
        yprimeiy = xdatixprime
    end do    /* Call wrdata with the revised arrays */
    call wrdata(pb2prime, xprime, yprime, kxmin, kx, kymin, ky, clabel, cunits, cfilt, cformt)
@#else
    call wrdata(pb2dat, xdat, ydat, kxmin, kx, kymin, ky, clabel, cunits, cfilt, cformt)
@#endif

    return
end

```

Write 2D array to HDF file. Borrowed this from the H collisional radiative code, irls.

⟨ Functions and Subroutines 1.2 ⟩ +=

```

subroutine wrdata(pb2dat, xdat, ydat, kxmin, kx, kymin, ky, clabel, cunits, cfilt, cformt)
    implicit none_f77
    @#if  $\neg$ HDF4
        po_common
    @#endif
        implicit none_f90

        integer kxmin, kx, ky, kymin    // Input
        character*(*) cunits, clabel, cfilt, cformt
        real xdatkx, ydatky, pb2datkx,ky
    @#if HDF4
        integer icount, iret, ix, iy, irank    // Local
        integer*4 idim2

        single_precision zdatic, zmax, zmin
        single_precision zxdatkx, zydatky, zdatakx*ky

        integer dssdast, dssdims, dssdisc, dssrang, dspdata
        external dssdast, dssdims, dssdisc, dssrang, dspdata, xerrab    /* Set dimension sizes */

        irank = 2
        idim1 = kx - kxmin + 1
        idim2 = ky - kymin + 1    /* Determine maximum and minimum; assign data to 1-D work array. */
        zmax = -const(1., 20)
        zmin = const(1., 20)
        icount = 0

        do iy = kymin, ky
            zydatiy = single(ydatiy)
            do ix = kxmin, kx
                zxdatix = single(xdatix)
                icount++
                zdatic = pb2datix,iy
                zmax = max(zmax, zdatic)
                zmin = min(zmin, zdatic)
                zdataicount = zdatic
            end do
        end do    /* HDF set calls */
        iret = dssdims(irank, idim)
        assert(iret  $\equiv$  0)
        iret = dssdisc(1, idim1, zxdat)
        assert(iret  $\equiv$  0)
        iret = dssdisc(2, idim2, zydat)
        assert(iret  $\equiv$  0)
        iret = dssdast(clabel, cunits, cformt, 'cartesian')
        assert(iret  $\equiv$  0)

        iret = dssrang(zmax, zmin)
        assert(iret  $\equiv$  0)    /* Write 2-D data */
        iret = dspdata(cfilt || '.hdf', 2, idim, zdata)
        assert(iret  $\equiv$  0)
    @#else    // !HDF

```

```

integer ix, iy, jx, jy     // Local

⟨ Memory allocation interface 0 ⟩

assert(po_nx ≡ kx - kxmin + 1)
assert(po_ny ≡ ky - kymin + 1)
po_num_dep_vars++
var_realloca(po_xscale)
var_realloca(po_yscale)
var_realloca(po_image_data)
var_realloca(po_image_labels)
var_realloca(po_image_units)
var_realloca(po_image_formats)
/* Define associated scales starting at kymin? (set auxiliary info: units, format, etc.) */
po_image_labelspo_num_dep_vars = clabel
po_image_unitspo_num_dep_vars = cunits
po_image_formatspo_num_dep_vars = cformat
do iy = kymin, ky
  jy = iy - kymin + 1
  po_yscalepo_num_dep_vars,jy = areal(iy)
  do ix = kxmin, kx
    jx = ix - kxmin + 1
    po_xscalepo_num_dep_vars,jx = areal(jx)
    po_image_datapo_num_dep_vars,jy,jx = pb2datix,iy
  end do
end do
@#endif
return
end

```

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〈Compute Signal 2〉 Used in sections 1.2 and 2.1.

〈Functions and Subroutines 1.2, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7〉 Used in section 1.1.

〈Memory allocation interface 0〉 Used in sections 2.7, 2.5, 2.4, 2.3, 2.2, 2.1, and 1.2.

COMMAND LINE: "fweave -f -i! -W[-ybs15000 -ykw800 -ytw40000 -j -n/
/Users/dstotler/degas2/src/postdetector.web".

WEB FILE: "/Users/dstotler/degas2/src/postdetector.web".

CHANGE FILE: (none).

GLOBAL LANGUAGE: FORTRAN.